

APPENDIX A

Below, we provide a detail explanation of the procedures employed in each of the models. All processes were performed on a GPU, and our Python code is available in our GitHub repository¹.

(1) Fine-tuned Transformers-based models

Here we used the Trainer class provided by Hugging Face, which offers an API for feature-complete training in PyTorch². The first step is downloading the pre-trained model. We tested two models derived from the BERT architecture: DeBERTaV3_{base}³ and DeBERTaV3_{large}⁴ (He et al., 2021a, 2021b). The primary difference between the two lies in their total number of parameters (435M for large and 184M for the base model) and the number of parameters in the embedding layer (131M vs. 98M). It is expected that the large model outperforms in tasks that require understanding intricate relationships, long-range dependencies, and fine-grained distinctions in the input data. On the other hand, the base model requires fewer computational resources for training and offers faster inference.

In this study, the annotated dataset consists of 1,000 headlines distributed across 10 classes. From the 1,000 headlines, a stratified sample of 20% was extracted to serve as the test set, while the remaining headlines were used to create stratified training sets with 200, 400, 600, and 800 data points.

Next, the training and test datasets are tokenized using the tokenizer provided by the respective models. The datasets are also converted into Hugging Face dataset object. The next step involves adjusting the training arguments and hyperparameters. It is important to note that there is no single definition of optimal hyperparameters, so we relied on a set of standard hyperparameters defined by Laurer and colleagues (2023), with the final configuration outlined within our Python script. The final step involves fine-tuning, employing the Trainer class.

¹ Available at: https://github.com/lffernandes08/ccr_codes

² Available at: https://huggingface.co/docs/transformers/main_classes/trainer

³ Available at: <https://huggingface.co/microsoft/deberta-v3-base>

⁴ Available at: <https://huggingface.co/microsoft/deberta-v3-large>

(2) Prompt-based models

We utilized the OpenAI API⁵, employing the gpt-3.5-turbo model to classify the test set of 200 headlines. At the time the experiments were conducted, it was the most capable and cost-effective model—according to the company—optimised for Chat Completions, which replicate the conversational capacity of ChatGPT. We used the standard hyperparameters recommended by the API, reducing randomness as much as possible (temperature=0.0).

The first experiment, which we called *GPT-zero*, was done without example headlines. The prompt used for this experiment was as follows: “You are a text classifier that will categorize headlines into 10 possible topics. The topics and their respective descriptions are as follows.” The prompt included the 10 classes and their descriptions (Table 1).

For the second experiment, which we called *GPT-few*, we used two headlines after the description of each topic. The introductory part of the prompt was slightly modified: “You are a text classifier that will categorize headlines into 10 possible topics. The topics and their respective descriptions are listed below, as well as two examples of headlines” (Table 2).

***GPT-zero* (only the name and class description)**

You are a text classifier that will categorize headlines into 10 possible topics. The topics and their respective descriptions are as follows:

1. Global Access to Vaccine

The headline addresses initiatives for equal access to COVID-19 vaccines, such as COVAX, and/or the need to combat inequity, ensure global distribution, and contain the disease, especially in low- and middle-income countries.

2. Science and Technology

The headline addresses the science behind vaccine development, including research, studies, safety and efficacy trials, emergency approval, and side effects.

3. Public Health Policies

The headline addresses vaccine mandates, vaccine passports, and/or other public health policies such as social distancing, quarantine, lockdowns, and mask mandates.

4. Vaccination Rollout/Campaign

⁵ Available at: <https://platform.openai.com/docs/overview>

The headline addresses vaccination rollout, campaigns, priority groups, vaccine deals, and/or the monitoring of vaccination rates.

5. Vaccine Hesitancy and Mis-/Disinformation

The headline addresses the circulation of disinformation and misinformation regarding vaccines, which contribute to vaccine hesitancy and reluctance.

6. Public Endorsement to Vaccine

The headline addresses public endorsement and incentive for vaccines, such as celebrities and artists getting vaccinated.

7. Institutional Affairs

The headline addresses institutional and governmental issues, political disputes, conflicts, political authorities, and export bans.

8. Problems in Vaccination

The headline addresses problems in vaccination, such as delays, fraud, disparities, vaccine shortages.

9. Public Perception of Vaccine

The headline addresses opinion polls and surveys on the public perception of vaccines and/or willingness to get vaccinated.

10. Economic Consequences

The headline addresses the impacts and benefits of vaccination on the economy.

Based on this instruction, classify the following headline, indicating the name of the topic:

Table 1: Prompt used in the *GPT-zero* model

***GPT-few* (name, class description, and 2 example headlines, totalling 20 headlines)**

You are a text classifier that will categorize headlines into 10 possible topics. The topics and their respective descriptions are listed below, as well as two examples of headlines:

1. Global Access to Vaccine

The headline addresses initiatives for equal access to COVID-19 vaccines, such as COVAX, and/or the need to combat inequity, ensure global distribution, and contain the disease, especially in low- and middle-income countries.

Examples:

“Vaccine equity key to defeating COVID-19 pandemic: global organization leaders”

“WHO official says COVID-19 vaccine deliveries to Africa intensified”

2. Science and Technology

The headline addresses the science behind vaccine development, including research, studies, safety and efficacy trials, emergency approval, and side effects.

Examples:

“Team behind Oxford COVID jab start final stage of malaria vaccine trials”

“Vaccine reduces hospitalization in Scotland 4 weeks after the 1st dose, shows research”

3. Public Health Policies

The headline addresses vaccine mandates, vaccine passports, and/or other public health policies such as social distancing, quarantine, lockdowns, and mask mandates.

Examples:

“Vaccine passports: a threat to liberty or necessary safeguard?”

“Mandatory vaccination needed in public service posts to eradicate virus”

4. Vaccination Rollout/Campaign

The headline addresses vaccination rollout, campaigns, priority groups, vaccine deals, and/or the monitoring of vaccination rates.

Examples:

“UK hopes for vaccine before Christmas”

“In Israel, people over 60 and medical workers will receive fourth vaccine doses”

5. Vaccine Hesitancy and Mis-/Disinformation

The headline addresses the circulation of disinformation and misinformation regarding vaccines, which contribute to vaccine hesitancy and reluctance.

Examples:

“Fauci doubts effectiveness of coronavirus vaccine in US due to anti-vaxxers”

“Vaccine hesitancy risks rising mortality”

6. Public Endorsement to Vaccine

The headline addresses public endorsement and incentive for vaccines, such as celebrities and artists getting vaccinated.

Examples:

“South Africa engages artists, athletes to ramp up vaccination”

“Billie Eilish: I would have died from COVID-19 if I hadn't been vaccinated”

7. Institutional Affairs

The headline addresses institutional and governmental issues, political disputes, conflicts, political authorities, and export bans.

Examples:

“Biden has blunt words for Republican governors on vaccines: ‘Get out of the way’”

“‘Sense of urgency’: premiers demand more COVID-19 vaccine information from Scott Morrison”

8. Problems in Vaccination

The headline addresses problems in vaccination, such as delays, fraud, disparities, vaccine shortages.

Examples:

“European countries turning to East for vaccines amid supply shortage”

“Delivery, distribution delays hinder COVID-19 vaccination drive in EU, US”

9. Public Perception of Vaccine

The headline addresses opinion polls and surveys on the public perception of vaccines and/or willingness to get vaccinated.

Examples:

“People from ethnic minorities less likely to accept COVID vaccine, says poll”

“Canadians divided over COVID-19 vaccination: survey”

10. Economic Consequences

The headline addresses the impacts and benefits of vaccination on the economy.

Examples:

“The I.M.F. sees a faster economic recovery as vaccines are deployed”

“World Bank projects global economy to grow by 4 pct in 2021 with widespread vaccination”

Based on this instruction, classify the following headline, indicating the name of the topic:

Table 2: Prompt used in the *GPT-few* model

(3) Off-the-shelf RTE-based models

Here we tested two popular models from Hugging Face, which were pre-trained and tailored for the Recognizing Textual Entailment (RTE) task. We refer to these models as *BART*⁶ and *RTE-general*⁷. One of the advantages of models based on the Transformers architecture is the availability of pipelines⁸, which are objects that abstract much of the complexity from the library. They offer a simple API and require only a few lines of code to perform various tasks.

For both models, we used the zero-shot-classification pipeline. The `candidate_labels` hyperparameter allows you to provide the model with labels for classification. In our case, this hyperparameter was defined with a list describing our classes (as presented in Table 3 of the main article). These classes function as hypotheses for the RTE task. Thus, for each of the headlines (`sequence_to_classify`), the model calculates the probability of entailment with the set of descriptions. The description with

⁶ Available at: <https://huggingface.co/facebook/bart-large-mnli>

⁷ Available at: <https://huggingface.co/MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli>

⁸ Available at: https://huggingface.co/docs/transformers/main_classes/pipelines

the highest probability is then assigned to the class to which the headline belongs (multi_label=False). The tests were conducted on the test set of 200 headlines.

(4) Fine-tuned Transformers-RTE-based models

The procedure used in these models is the same as that of the first ones (Fine-tuned Transformers-based models), with the addition of a data formatting step, precisely for including RTE classification (Laurer et al., 2023). In the training dataset, for each headline of a specific class, a column is added with the class's description (which functions as the RTE hypothesis), and another column, "True", indicating that the description entails a headline. Additionally, for each headline, a random description (hypothesis) is included so that the model can identify false headline-description (text-hypothesis) pairs. This effectively doubles the size of the training dataset.

From that point onward, the protocol follows the previous one: downloading pre-trained models (here we also use a base model⁹ and a large model¹⁰, derived from the BERT architecture), tokenization, setting hyperparameters, and fine-tuning using the Trainer class. The procedure was also repeated for each model (base and large) with different training datasets (200, 400, 600, and 800 data points).

References

- He, P., Gao, J., & Chen, W. (2021a). DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2111.09543>
- He, P., Liu, X., Gao, J., & Chen, W. (2021b). DeBERTa: Decoding-enhanced BERT with Disentangled Attention. *International Conference on Learning Representations*. <https://openreview.net/pdf?id=XPZlaotutsD>
- Laurer, M., Van Atteveldt, W., Casas, A., & Welbers, K. (2023). Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI. *Political Analysis*, 1–17. <https://doi.org/10.1017/pan.2023.20>

⁹ Available at: <https://huggingface.co/MoritzLaurer/DeBERTa-v3-base-mnli-fever-docnli-ling-2c>

¹⁰ Available at: <https://huggingface.co/MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli>